

Resilience of southern Yukon boreal forests to spruce beetle outbreaks

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ABSTRACT

Changed disturbance regimes could drive biome-level shifts in vegetation structure and have cascading societal consequences. Expanding and intensifying bark beetle outbreaks pose a risk of major changes to large regions of boreal forests in North America. We evaluated the resilience of boreal forests in southwestern Yukon to an unprecedented spruce beetle (*Dendroctonus rufipennis* Kirby [Coleoptera: Curculionidae]) outbreak using data collected from permanent plots, which were measured repeatedly, in 21 stands. We determined basal area (m^2/ha) and density (stems/ha) of canopy trees (which are most vulnerable to beetles), density of advance regeneration, and growth response to the outbreak using ring widths from over 800 trees ranging in size from small advance regeneration to canopy trees. Beetle-related reductions in tree canopy basal area averaged 52% across all stands but there was considerable variability among stands. About 68% of variation in basal area reductions among stands was explained by variability in pre-outbreak canopy tree basal area ($\beta = 1.1$; $p = 0.0021$) and a climatic moisture deficit index ($\beta = 0.5$; $p = 0.0015$). Although the percentage of white spruce (*Picea glauca* (Moench) Voss) in the canopy was reduced in some stands, by 2–29%, white spruce remained the dominant canopy tree species. Almost all stands had more than adequate advance regeneration density (average = 5 448 stems/ha) in 2016 to replace canopy trees killed by beetles. Variability in advance regeneration among stands could partly be explained by pre-outbreak tree canopy basal area ($\beta = -0.4$; $p = 0.0018$), a climate moisture deficit index ($\beta = -0.3$; $p = 0.0004$), and shrub per cent cover ($\beta = -0.3$; $p = 0.0003$), as well as, humus depth ($\beta = 544$; $p = 0.0186$). Over the last two decades, spruce trees, of all size classes, exhibited substantial increases in mean annual radial growth increment, beginning 4–5 years after the start of the outbreak. The proportion of trees in a stand with at least a 50% increase in radial growth increment varied significantly with canopy tree basal area at the end of the outbreak ($\beta = -0.12$; $p < 0.0001$), climatic moisture deficit ($\beta = -0.02$; $p < 0.0196$), and humus depth ($\beta = 0.26$, $p < 0.0001$). Our findings indicate high response diversity to disturbance and suggest that forests of southwestern Yukon have high resilience to the recent spruce beetle outbreak because of the large number of surviving canopy trees, the abundant advance regeneration of spruce that can replace beetle-killed trees, and the increased growth of surviving trees.

1. Introduction

Boreal forests cover an extensive area of the northern hemisphere and provide important ecosystem services that support societal well-being at local, regional and global scales: wood supply, biological diversity, water quality, climate regulation, cultural inspiration, and recreation (Iverson et al., 2018). Disturbances play a key role in the dynamics of healthy boreal forests, re-initiating or accelerating successional sequences and maintaining complex forest structures that: (i) make boreal forests resilient to subsequent expected, or unforeseen, disturbances and, (ii) sustain ecosystem services (Levin, 2005; Peutmann et al., 2008). However, global change, particularly a

warming climate, is altering boreal environments and changing the characteristics of forest disturbance regimes (Price et al., 2013; Gauthier et al., 2015; Seidl et al., 2016; Aoki et al., 2018). Millar and Stephenson (2015) suggest that climate change is initiating an era of “mega-disturbance”, whereby more extensive and frequent disturbances, including the occurrence of novel disturbances, may push forests over thresholds that exhaust resilience. Shifts to more severe, and possibly novel, disturbance regimes will alter forest structure and provide opportunities for post-disturbance vegetation to reorganize in very different ways, along trajectories that could lead to fundamentally different ecosystem states (Seidl et al., 2016). While reductions in tree cover are expected in some parts of the boreal forest, in other regions

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more severe disturbances are expected to drive boreal forest transitions to deciduous broadleaf forests, shrublands or grasslands by the end of the century (Chapin et al., 2003; Chapin et al., 2010; Johnstone et al., 2010; Pelz and Smith, 2013; Scheffer et al., 2012; Gauthier et al., 2015). This would have substantial implications for the continued flow of ecosystem services, especially if such ecosystem transformations occur suddenly.

Fire disturbance is the most important driver of forest structure and dynamics in many regions of the boreal forest (Johnson, 1992; Payette, 1992; Bergeron et al., 2004) but in others, insect disturbances are of equal or greater significance (Bergeron, 1998; Candau et al., 2018). Spruce budworm (*Choristoneura fumiferana* Clemens [Lepidoptera: Tortricidae]) outbreaks have caused widespread tree mortality in boreal forests of eastern Canada, especially where fires are infrequent (Campbell et al., 2008; Shorohova et al., 2011). In contrast, bark beetle outbreaks have historically had limited impacts in boreal forests of North America, but outbreaks are intensifying and expanding (Berg et al., 2006; Cullingham et al., 2011; Fuentealba et al., 2013).

Spruce beetle (*Dendroctonus rufipennis* Kirby [Coleoptera: Curculionidae]) outbreaks have been a major disturbance in many forests of western North America, from the southern Rocky Mountains (Eisenhart and Veblen, 2000; DeRose and Long, 2012), through British Columbia (Zhang et al., 1999), to Alaska (Doak, 2004; Berg et al., 2006; Csank et al., 2016; Hansen et al., 2016). Outbreaks have been expanding into, or intensifying, in the boreal forest, as warming climates hasten beetle development, increase beetle over-wintering survival, and compromise host tree resistance to infestation (Berg et al., 2006; Raffa et al., 2008; Bentz et al., 2010; Cullingham et al., 2011; Chavardes et al., 2012). Given that spruce beetle outbreaks have the potential to kill most mature *Picea* in a stand, as has been observed in the southern Rocky Mountains of Colorado and Utah (Veblen et al., 1991; DeRose and Long, 2010), altered spruce beetle disturbances could be major agents of socio-ecological change in boreal forests (Morris et al., 2017). This threat of change to boreal forests is heightened by the perception that more severe outbreaks increases risk of fire disturbance (Morris et al., 2017).

Like many native insect disturbances, bark beetle disturbances are species-specific and outbreaks of *Dendroctonus* spp. typically only kill large overstory trees, leaving the understory mostly intact (e.g., Astrup et al., 2008; Pelz and Smith, 2012; Jenkins et al., 2014; Campbell and Antos, 2015). Consequently, these bark beetle outbreaks can produce fundamentally different post-disturbance stand development trajectories than fires, which kill small and large trees of any species (Veblen et al., 1991; Campbell and Antos, 2015; Paudel et al., 2015). The formation of a new canopy following bark beetle-caused mortality can result largely from growth of understory trees (Nigh et al., 2008; Axelson et al., 2009; Vyse et al., 2009; Hawkins et al., 2012, 2013; Amoroso et al., 2013). In addition, many canopy trees frequently survive an outbreak in some regions, including host species, and these can make a major contribution to canopy recovery (Hawkins et al., 2012; Amoroso et al., 2013). Like mountain pine beetle outbreaks on pines, spruce beetle outbreaks change light and competitive environments, allowing for rapid growth of advance regeneration, which forms a new canopy along with surviving trees. Shade-tolerant Engelmann spruce (*Picea engelmannii* Parry ex Engelm.) and subalpine fir (*Abies lasiocarpa* (Hook.) Nutt.) form large seedling banks (advance regeneration) in many spruce-fir forests (e.g., Antos et al., 2000; DeRose and Long, 2010), providing substantial potential for canopy redevelopment without the need for new tree establishment following beetle mortality of canopy spruce. In contrast, new establishment is typically required for canopy recovery following fires.

High-latitude ecosystems are expected to face increasing tests of their resilience to disturbances under a changing climate (IPCC, 2014; Reyher et al., 2015). An improved understanding of how boreal forests are responding to novel or intensifying disturbances is a key element to forecasting future landscape changes and guiding management actions

to mitigate or adapt to socio-ecological effects of altered disturbance regimes (Millar and Stephenson, 2015; Seidl et al., 2015; Gauthier et al., 2015). In this study, we assess the resilience of boreal forests in southwestern Yukon to intensifying spruce beetle outbreaks, determining whether, or not, these forests are on a recovery trajectory, or moving towards a different state. To do this, we used data from 21 stands that were sampled three times after the outbreak started, and reconstructed pre-outbreak canopy conditions from living and dead trees present at the first sampling. Specifically, we quantify three factors that collectively influence the capacity of these forests to reorganize and maintain similar forest structure following disturbance: (1) per cent reduction in canopy tree density and basal area (i.e., disturbance severity); (2) the amounts, size distribution, and species composition of advance regeneration; and (3) the growth response of advance regeneration and residual canopy trees to the outbreak using ring width measurements.

2. Methods

2.1. Study area

We studied valley-bottom, boreal forests of southwestern Yukon, Canada, near Haines Junction (Fig. 1) between latitudes N60.298 and N61.157. Permafrost occurs in scattered areas throughout the valley, mostly on north slopes and in areas of peat accumulation or poor drainage (Lewkowicz et al., 2012). The study area lies in the rain-shadow of the St. Elias Mountains and occurs at the confluence of cold, dry air masses from the arctic and warm, moist maritime air masses that cross over the St. Elias Mountains from the Pacific Ocean. This modified boreal climate is characterized by a mean annual temperature (MAT) of -2.5°C ($+/-0.6^{\circ}\text{C}$) (supplement Fig. S1). The area has 684 growing degree days (i.e., degree-days above 5°C) and a frost-free period of 59 days. Relative to most other parts of the Canadian boreal forest, the study region is semi-arid, with only 312 mm of precipitation annually, half of this falling as summer rain. Projected climate change for this region in the next 80 years is substantial (supplement Fig. S1). Models indicate mean annual temperature increases between 2 and 6°C and about a 50% increase in growing season length. Projected increases in annual precipitation may not be enough to offset the effects of increased temperatures; the climatic moisture deficit (CMD), an indicator of drought, is projected to increase by $\sim 22\text{--}26\%$ (Wang et al., 2016).

In the lowland valley bottoms (760–1080 m elevation), 86% of the landscape is comprised of stands dominated by white spruce (*Picea glauca* (Moench) Voss), the only coniferous tree species in this region of Yukon (Krebs et al., 2001). White spruce forms both pure and mixed-species stands, which can contain trembling aspen (*Populus tremuloides* Michx.), balsam poplar (*P. balsamifera*) and Scouler willow (*Salix scouleriana* Barratt). Relative species abundance in mixed stands depends on edaphic conditions and time since last fire disturbance. Stands dominated by trembling aspen comprise 13% of the landscape (Krebs et al., 2001), and typically occur on south slopes and at higher elevations than spruce stands.

A spruce beetle outbreak occurred between 1932 and 1942 (Berg et al., 2006) and a more severe outbreak occurred recently (between 1994 and 2006), which was unprecedented in spatial scale affecting most spruce forests in the study area (Yukon Government, 1994–2014) (Figs. 1 and 2). About every 10 years, snowshoe hares cause substantial damage to spruce regeneration (Krebs et al., 2001). Other biotic disturbances of spruce-dominated forests – such as spruce budworm (*Choristoneura* spp.) and tomentosus root disease – cause relatively little tree mortality (Yukon Government, 2002–2017). Fires have had less impact within the study area than in other regions of Yukon, or western Canada, with fire return intervals ranging from 200 to 500 years (Hawkes, 1983; Francis, 1996; de Groot et al., 2013). Human land use activities are limited; no large-scale commercial logging operations occur within the study area. Major drought periods for this region

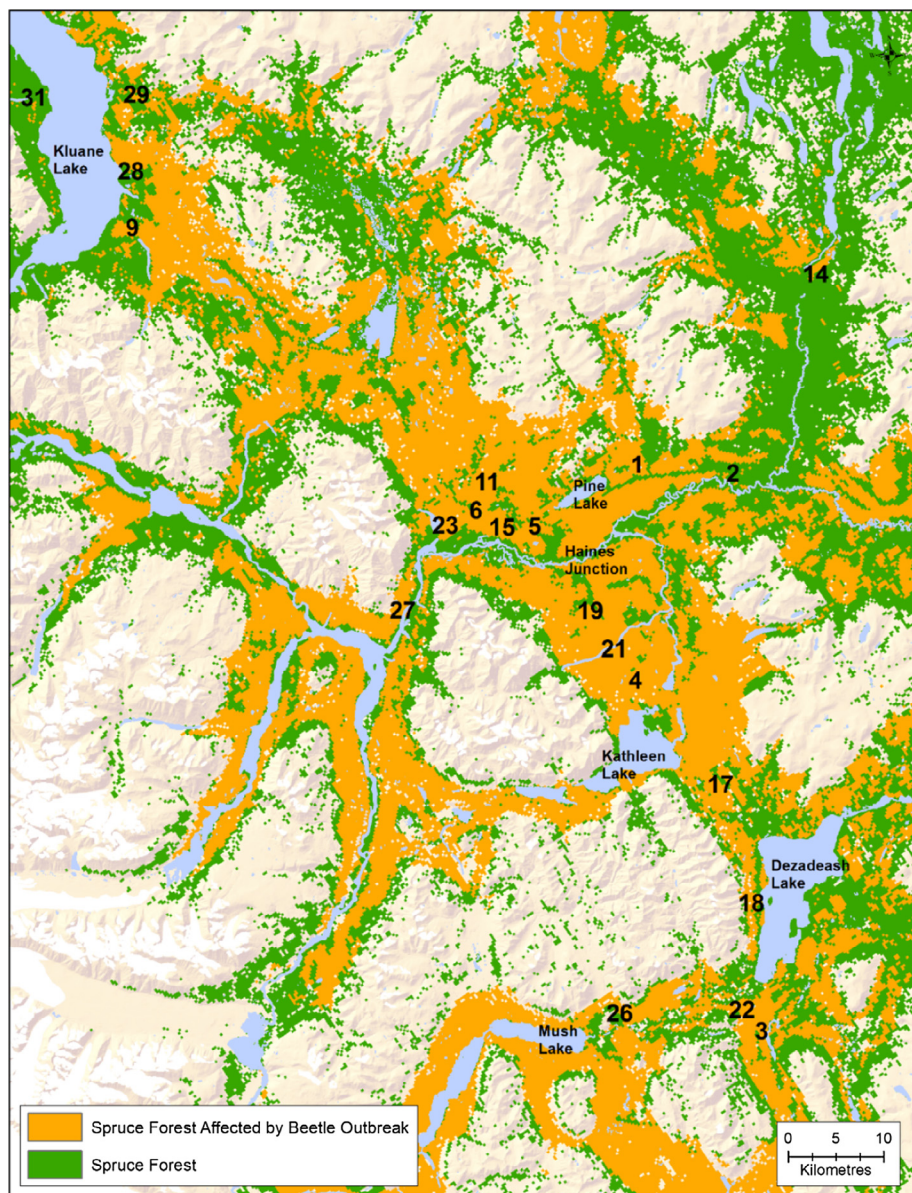


Fig. 1. Location of 21 stands sampled in southwestern Yukon and spatial extent of the last spruce beetle outbreak (1994–2006) determined from aerial survey data. Yellow includes areas with trace (< 1% mortality) to severe (> 50% mortality) outbreak intensity.

occurred from 1925–28, 1940–42, 1946–48, 1956–58 and 1993–95 (supplement Fig. S1).

2.2. Field sampling and tree-ring measurement

During the summers of 2000 and 2002, while the spruce beetle outbreak continued to expand in the study area, permanent sample plots were established in 21 stands, using a stratified random sampling design (Garbutt et al., 2006). At this time, all stands had on-going infestations or signs of infestation in previous years. Four plots (25×4 m) were sampled in each stand to assess: the number of trees infested and killed by spruce beetle, forest structure, tree species composition, understory species composition and cover, site characteristics (elevation, slope inclination, etc.), and soil properties (see Garbutt et al., 2006 for plot establishment and detailed sampling protocols). The stands were re-measured in 2010 and 2016 to quantify post-outbreak stand attributes and assess trajectories of forest recovery. All living and standing dead trees were tallied in each plot. Breast height diameter (dbh) was measured for all trees ≥ 1.3 m tall; trees

0.3–1.3 m tall were counted. Seedlings (< 0.3 m tall) were counted in 2016 and the substrate on which they were growing (moss, humus, mineral soil) was recorded. Also in 2016, increment cores, and basal disks from small trees, were collected for a sample of trees in each plot to determine radial growth responses of surviving trees to the recent beetle outbreak. Trees were sampled to reflect the relative abundance of canopy trees (> 10 cm dbh), subcanopy trees (0.1–10 cm dbh), and saplings (0.3–1.3 m tall) in each plot. An average of 45 increment cores and basal disks were collected in each stand. Cores and disks were collected as close to the bottom of trees as possible.

All increment cores and basal disks were prepared in the laboratory following standard dendrochronology procedures (Swetnam et al., 1985). After sanding and mounting of cores, tree rings were visually cross-dated within plots (*sensu* Yamaguchi, 1991). Cores and disks were scanned using a flatbed scanner and the ring widths on the images measured to the nearest 0.01 mm using a WinDendro™ or a Velmetax TA Unislide Measuring System. Cross sections with very tight rings were scanned using a Hitachi S-3400N scanning electron microscope (SEM) and measured using WinDendro™. The program COFECHA (Holmes,

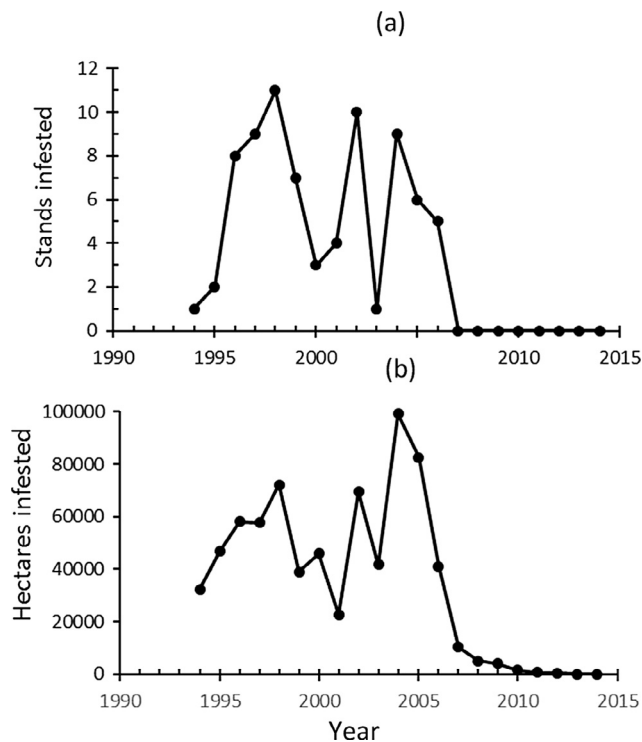


Fig. 2. Annual aerial survey detection of the number of our sample stands infested each year (a) and the number of hectares of spruce forest infested (b) by bark beetles over the entire study area in southwestern Yukon.

1983) was used to assess the accuracy of the date assigned to each tree ring. For the time period of interest (1980–2015), we could accurately assign dates to tree rings on 93% of the samples; the other 7% were excluded from further analyses.

2.3. Data summary and analysis

To describe the annual progression of the spruce beetle outbreak over our entire study area, and in each stand, we queried digitized aerial survey maps identifying polygons of infested forest and recording severity of the infestations in these polygons as light, moderate, or severe (Yukon Government, 1994–2014). We calculated the annual area (ha) infested by beetles for the entire study area and graphed a time series of outbreak extent. We also overlaid stand locations (surrounded by a 200 m buffer) onto the aerial survey maps to determine infestation timing (start year and duration) for each stand and to obtain an estimate of outbreak severity.

To quantify beetle outbreak severity and the degree of subsequent stand recovery we needed an estimate of the pre-disturbance canopy tree density and basal area for the stands. The first sampling was conducted after the initiation of the outbreak and thus could not be used directly for this purpose, but this sampling was conducted soon enough to confidently determine stand conditions prior to tree mortality from the beetle. We reconstructed the density and basal area of canopy trees before the outbreak by combining the number of living trees at the time of sampling with the number of trees recently killed by spruce beetle (i.e., standing dead or recently fallen trees with signs of entrance holes and beetle galleries). Because mortality due to other causes is low, these values should closely approximate the density and basal area of living canopy trees at the beginning of the outbreak.

We evaluated the potential for stands to recover canopy tree density and basal area by determining the abundance and species composition of advance regeneration, which we defined as all small trees that established prior to the start of the spruce beetle outbreak. We considered advance regeneration in two size categories – saplings (0.3–1.3 m tall)

and subcanopy trees (> 1.3 m tall but < 10 cm dbh). Both have substantial potential for increasing in growth and entering the canopy following loss of previous canopy trees.

Several approaches were used to evaluate changes in tree growth related to the beetle outbreak. We characterized time series of mean annual radial growth increment (i.e., tree ring width) from 1980 to 2015, using a large sample of surviving trees in each of three size classes (canopy trees, $n = 162$; subcanopy trees, $n = 441$; and saplings, $n = 200$). To identify years of significant transitions (i.e., breakpoints) in mean tree growth, we fit piecewise linear regression models to the time series using the ‘segmented’ R package (Muggeo, 2017). To gain insight about the influence of climate change on increases in mean tree growth increment, which we presumed to be caused largely by the beetle outbreak, we tested for a relationship between the increasing growth increment (i.e., the regression segment after the breakpoint identified by piecewise regression) and degree days > 5 °C, the climate variable showing the strongest directional change from 1998 to 2015 (supplement Fig. S1). We also examined variability in individual tree responses to the beetle outbreak by comparing growth before and after the outbreak. Changes in growth for each of the three size classes was analyzed separately as we expected them to show different responses to tree mortality caused by the outbreak.

To determine what factors contribute to variation in outbreak severity and recovery potential among stands, we undertook multiple regression analyses on three dependant variables: 1. per cent reduction in canopy tree basal area at the end of the outbreak (from plot measurements in 2010); 2. density of advance regeneration (stems/ha) surveyed in 2016, by size class, and 3. proportion of subcanopy trees with at least a 50% increase in growth after the outbreak (2010–2015) compared to pre-outbreak (1980–1985). Previous studies show tree radial growth increases of at least 50% effectively identify growth responses due to canopy thinning by bark beetle outbreaks (Berg et al., 2006; Campbell et al., 2007; Amoroso et al., 2013). We conducted statistical analyses of growth response on subcanopy tree data because there were insufficient numbers of saplings, or surviving canopy trees, with core or basal disk samples in many stands. We tested the effects of biologically meaningful sets of independent variables for each dependent variable (supplement Table S1). Knowledge of factors contributing to spruce beetle outbreak occurrence and severity (Doak, 2004; Berg et al., 2006; Raffa et al., 2008; Bakaj et al., 2016) and post-outbreak stand development, including regeneration abundance and tree growth response to insect disturbances (Veblen et al., 1991; Franklin et al., 2007; Campbell and Antos, 2015; Maclean, 2016) guided variable selection. Examples of variables examined include pre-outbreak canopy tree basal area, percentage of spruce in the canopy, growing degree days, extreme winter temperature, climatic moisture deficit, forest floor depth (a proxy for forest productivity), stand age, average canopy tree dbh, and seed bed abundance (moss, mineral soil, humus) (see supplement Table S1 for variable details). We used the least absolute shrinkage and selection operator (LASSO) procedure (Tibshirani, 1996) – which outperforms subset variable selection procedures for multiple regression analyses with collinear variables and small N (Dahlgren, 2010) – to inform the selection of explanatory variables to include in multiple regression models. We used R (R Development Core Team, 2016) to build generalized linear regression models and the ‘glmnet’ R package (Friedman et al., 2018) to implement the LASSO procedure. Basal area reduction (%) and advance regeneration density (stems/ha) were modelled with a gaussian distribution while the proportion of subcanopy trees with at least a 50% increase in growth increment was modelled with a binomial distribution. Standard graphics of model residuals and statistical tests were used to determine if the final models satisfied the assumptions of multiple linear regression.

3. Results

3.1. Spruce beetle outbreak patterns

Aerial surveys detected spruce beetle infestations between 1994 and 2012 in our study area (Fig. 2). At the peak of the outbreak in 2004, beetles infested nearly 100,000 ha of spruce forest (Fig. 2). After infestations were first detected in 1994, the annual area affected increased until 1998. Between 1999 and 2001, the area decreased but increased again in 2002. The outbreak began to collapse in 2006 and no infestations were recorded after 2008. The outbreak pattern in the 21 stands was almost identical to the landscape-level outbreak pattern observed across all spruce forests in southwestern Yukon. Aerial surveys detected beetle infestations in our stands between 1995 and 2006, with the highest number of infested stands occurring in 1998, 2002, and 2004 (Fig. 2). The number of years an infestation was detected in each stand ranged from none to seven. Annual infestations were recorded as light to moderate in all stands except two (22 and 26), which had severe infestations in 1998 and 2004.

3.2. Impact of spruce beetle outbreak on forest canopy structure

We documented substantial reductions in average canopy tree basal area (42%) (m^2/ha) and canopy tree density (stems/ha) (32%) across all 21 stands from pre-outbreak to the first stand measurement (Table 1). However, there was considerable variability in the impact of the beetle outbreak among stands, with reductions in basal area from 2.1 to 87% (Fig. 3, Table 1). From the first measurement to the second

in 2010, basal area declined an additional 5–51% in 13 of the 21 stands but began to increase in the eight other stands; similar patterns of change were observed for canopy tree density (Fig. 3, Table 1). Canopy tree basal area and density increased in four stands by 2010, but recovery was slow. Between 2010 and 2016, canopy tree basal area and density increased in 10 of the 21 stands and rates of gain were quite fast in some stands. However, at the last measurement in 2016, or 10 years after the end of the outbreak, only five of the 21 stands had attained pre-outbreak canopy tree basal area or density, and many stands still had greatly reduced canopy tree density and basal area.

The species composition of the forest canopy changed very little over the measurement period (Fig. 3, Table 1). The canopy in 12 stands was comprised of 100% white spruce before the outbreak and at each measurement time thereafter. Among mixed species stands, the percentage of white spruce in the canopy decreased in six stands by 2016, with trembling aspen 16–19% more prominent after the outbreak. In the remaining mixed stands, the percentage of spruce in the canopy increased in two stands by 2016 (10 and 14%) but remained unchanged in one stand. Overall, the outbreak had a small net effect on the tree species composition of the forest canopy.

3.3. Advance regeneration

In 2016, advance regeneration (trees that established before the outbreak started) was abundant in all stands but varied considerably among stands ranging from 700 stems/ha to just over 11 000 stems/ha (Table 1). On average, the density of advance regeneration in 2016 (5448/ha) was much higher than the pre-outbreak density of canopy

Table 1

Forest structure and composition for each of 21 stands sampled in the southwestern boreal forests of Yukon. Pre-outbreak canopy tree density (stems/ha), basal area (m^2/ha), % white spruce were estimated from 2000 to 2002 measurements and stands were reassessed in 2010 and 2016. Advance regeneration counts were done in 2016 and include all stems 0.3 m to breast height (1.3 m) and stems with breast height diameters up to 10 cm.

Stand	Canopy trees												Advance ³ Regeneration 2016	
	Pre-outbreak canopy			2000			2010			2016			Density (stems/ha)	% Sw
	Tree	Basal	Area (m ² /ha)	Tree	Basal	Area (m ² /ha)	Tree	Basal	Area (m ² /ha)	Tree	Basal	Area (m ² /ha)		
Density (stems/ha)	% Sw	Area (m ² /ha)	Density (stems/ha)	% Sw	Area (m ² /ha)	Density (stems/ha)	% Sw	Area (m ² /ha)	Density (stems/ha)	% Sw	Area (m ² /ha)	Density (stems/ha)	% Sw	
1	1575	98	44.9	1250	98	35.2	625	96	17.2	600	96	15.4	4625	96.2
2	1000	88	24.1	975	87	23.6	875	97	16.5	1125	98	21.3	3575	60.8
3	1850	84	48.7	1025	71	26.6	625	52	14.4	400	55	7.5	3475	75.5
4	1200	77	25.5	1050	74	20.1	1000	70	17.7	750	70	11.9	11,375	79.3
5	1800	100	46.1	1075	100	24.1	775	100	15	875	100	16.6	10,175	100
6	740	100	28	400	100	12	240	100	8.9	550	100	15.7	4475	100
9	740	100	16.8	680	100	13.6	640	100	12.8	925	100	16.9	7025	100
11	1725	97	27	1450	97	21.6	1375	96	24.2	1925	97	31.7	2600	100
14	1425	100	31.3	1000	100	22.4	1150	100	27.8	1175	100	29.5	8950	100
15	1075	100	45.3	800	100	19	825	100	23.2	1067	100	28.4	2300	100
17	825	73	30.2	575	61	19.3	525	67	17.8	667	87	11.6	8825	45.9
18	850	71	26.9	725	66	21.8	475	47	16.1	275	52	9.2	4350	71.8
19	1450	100	26.1	925	100	13.5	775	100	12.4	900	100	15	9733	96.2
21	1850	69	54.7	1275	59	36.1	800	56	22.6	400	53	9.4	5500	41.8
22	850	100	30.3	375	100	11.7	200	100	6.5	200	100	7.1	925	97.3
23	700	100	42.6	350	100	14.6	275	100	13.8	433	100	21.7	700	100
26	700	68	17.3	650	65	15.9	575	57	14.3	400	60	5.7	7200	67.6
27	750	100	34.7	325	100	11.8	275	100	12.7	375	100	15.9	1275	100
28	1325	100	39.5	325	100	5.1	500	100	8.3	625	100	13.1	5425	100
29	675	100	8.9	625	100	8.5	650	100	8.9	733	100	9.6	6033	100
31	850	100	43.4	625	100	31.6	475	100	23	475	100	24.2	5875	100
Mean	1141	92	33	785	89	19	652	87	16	708	89	16	5448	87
Min	675	68	9	325	59	5	200	47	7	200	52	6	700	42
Max	1850	100	55	1450	100	36	1375	100	28	1925	100	32	11,375	100

1. We reconstructed pre-outbreak forest canopy density and basal area by combining trees recently killed by the spruce beetle outbreak with living trees at the time of plot establishment in 2000/02.

2. Stands that did not have 100% white spruce contained deciduous species such as aspen, balsam poplar and willow.

3. Advance regeneration includes small trees establishing before the start of the outbreak; establishment dates were determined from tree rings.

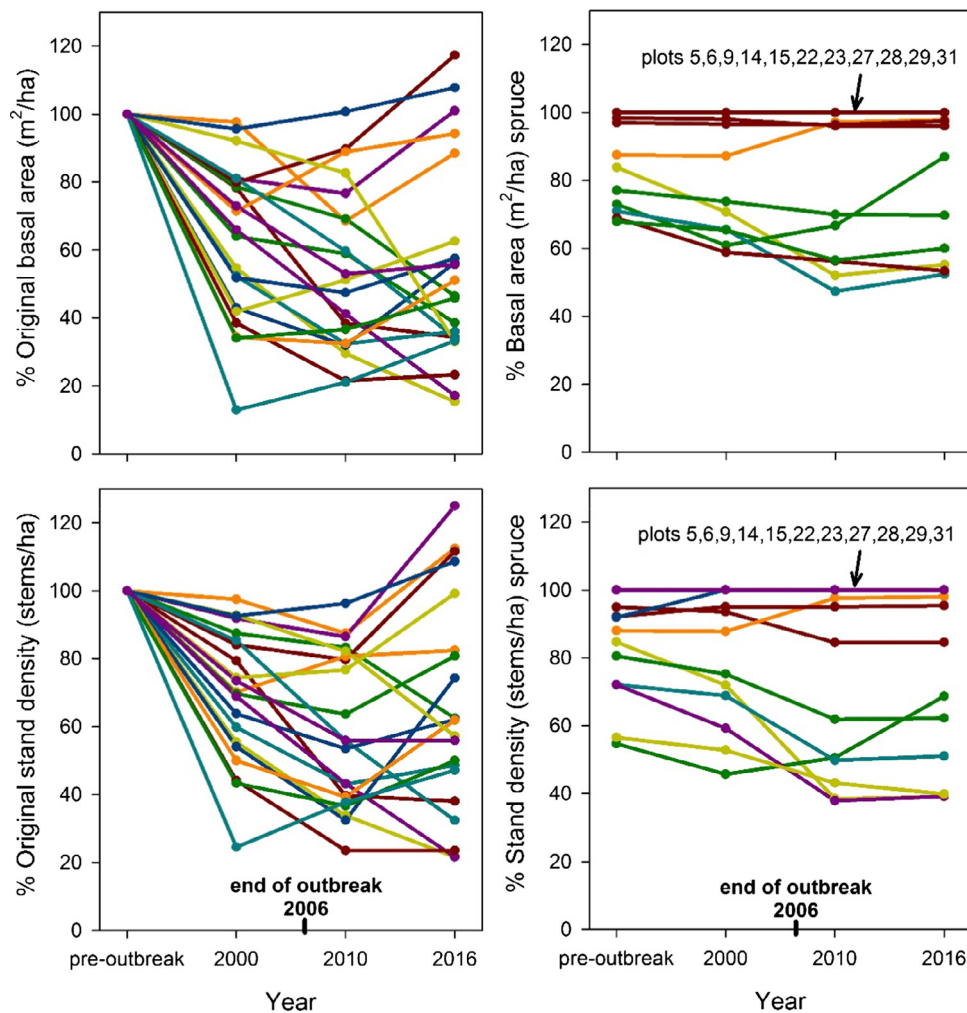


Fig. 3. Changes in basal area and density of living canopy trees at three sampling periods (2000/2002, 2010, and 2016) relative to pre-outbreak values for 21 stands in southwestern Yukon. Panels on the left are total canopy tree basal area or density, and panels on the right show per cent of canopy tree basal area or density contributed by white spruce. Pre-outbreak values were reconstructed from living and recently dead trees present during the 2000/2002 sampling period (see Table 1 for values for basal area (m^2/ha) and density (stems/ha) at each sampling period).

Table 2

Mean density (stems/ha) and standard deviation (SD) of advance regeneration abundance (stems/ha), by species, and size class for 21 stands sampled in southwestern Yukon in 2016. Advance regeneration includes regeneration establishing before the start of the beetle outbreak in 1994.

Species	Large subcanopy (5–10 cm dbh)		Small subcanopy (0–5 cm)		Saplings (0.3–1.3 m tall)	
	Mean	SD	Mean	SD	Mean	SD
white spruce	769	724	1453	1148	2431	1947
trembling aspen	28	71	195	368	478	1000
balsam poplar	2	11	33	137	20	73
willow	4	15	20	70	4	16
All species	803		1701		2933	

trees (1141/ha) (Table 1). Moreover, the percentage of spruce among advance regeneration (87%) was only slightly lower than the percentage of spruce in the canopy prior to disturbance (92%). Similar to the forest canopy, advance regeneration was comprised of 100% spruce in just over half of the stands (Table 1). In the remaining stands, 0–58% of the regeneration was comprised of other species (Table 1), including trembling aspen, balsam poplar and willows (Table 2). White spruce saplings were about two times more abundant than small subcanopy trees and 3 times more abundant than larger subcanopy trees (Table 2).

White spruce regeneration was about seven times more abundant than trembling aspen, but the distribution of aspen among sizes classes was similar to spruce (Table 2).

3.4. Survivor growth

After the beetle outbreak, mean radial growth increased similarly among trees of all size classes: canopy, subcanopy, and saplings (Fig. 4). Piecewise linear regression analyses of time series indicated statistically significant increases in mean radial growth increment (mm/yr) began between 1998 and 1999 (Fig. 4), four to five years after the start of the beetle outbreak. Even though growing degree days also gradually increased from 1998 and 2015, there was no significant linear relationship between growth increment and growing season length ($F = 1.8$, $p = 0.1936$; $R^2 = 0.05$). While growth increment of saplings and subcanopy trees continued to increase from 1998 to 2015, growth of canopy trees stabilized over the last decade. Not only did mean growth increase following the outbreak but variation in annual growth increment also increased for all size classes (Fig. 4).

To further explore changes in growth related to the outbreak, we also examined growth responses of individual trees, by size class, comparing growth rates before the outbreak (1980–1985) to growth after the outbreak (2010–2015) (Fig. 5). There was a significant increase in mean radial growth increment after the outbreak among all

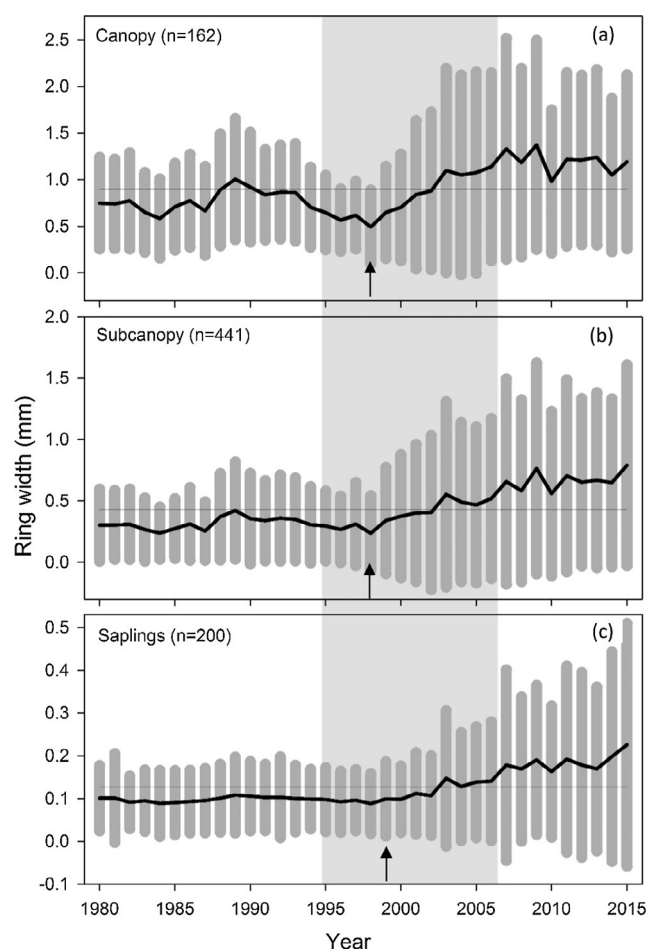


Fig. 4. Mean annual ring width (mm) and standard deviation (grey bars) from 1980 to 2015 for (a) canopy trees, (b) subcanopy trees, and (c) saplings surviving the last spruce beetle outbreak in 21 stands of southwestern Yukon. The light grey horizontal line is mean ring width for the entire period. Grey shading indicates the spruce beetle outbreak period and the black arrows statistically significant breakpoints in growth trend as determined by piecewise linear regression (p -values for a score statistic test of slope differences: canopy ($p < 0.0001$), subcanopy ($p = 0.0005$), saplings ($p < 0.0001$); segmented R package, Muggeo, 2017).

tree size classes (Table 3) and increases in growth of subcanopy trees were two times greater than the mean increase for canopy trees and a third more than increases in growth of saplings (Table 3). While most trees exhibited increased growth after the outbreak, the growth of others changed little or decreased (Fig. 5). About half the surviving spruce in all canopy classes exhibited increases in radial growth of at least 50% but very few saplings had such extreme increases in growth as observed among canopy and subcanopy trees. Although sapling growth increased greatly following the outbreak, growth was still very slow (0.2 mm/yr) compared to the growth of subcanopy and canopy trees: 0.6 mm/yr and 1.2 mm/year, respectively (Figs. 4 and 5). For all size classes, growth increment before the outbreak was not a good indicator of growth after the outbreak; fast growers did not necessarily increase in growth while some of the very slow growers increased substantially (Fig. 5).

3.5. Variability across stands

The best multiple linear regression model of relationships between per cent reduction in tree canopy basal area in 2010 (i.e., beetle outbreak severity) and stand variables, reduced residual deviance compared to the null model, by 68% (Table 4). We found a significant

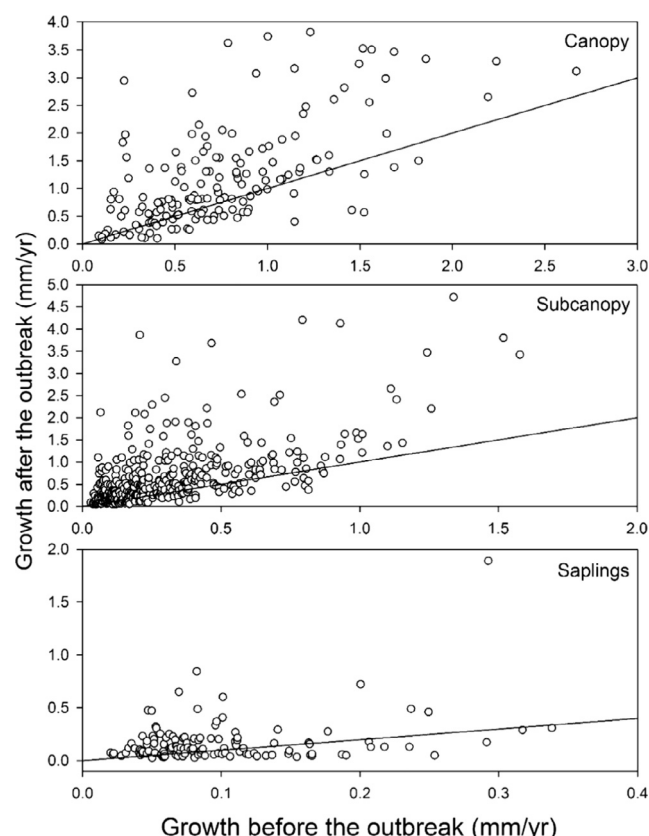


Fig. 5. Comparisons of tree growth before and after the outbreak. Mean growth (mm/yr) after the outbreak (2010–2015) vs. mean growth (mm/yr) prior to the outbreak (1980–1985) for individuals of three size categories: canopy, subcanopy and saplings. The black line is a reference line indicating no change in growth (i.e., the same radial growth before and after the outbreak).

Table 3

Growth increment and increase (%) among canopy and subcanopy trees and saplings following a spruce beetle outbreak in southwestern Yukon. Per cent increases were calculated comparing growth (mm/yr) before the outbreak (1980–1985) to growth after the outbreak (2010–2015) across all trees in each size class.

	N	Average annual increment (mm/yr)		% growth increase		
		1980–1985	2010–2015 ¹	Mean ²	Median	Std. Dev.
Canopy	163	0.688	1.182	106	52	191
Subcanopy	422	0.276	0.655	211	95	368
Saplings	131	0.094	0.177	141	51	282

¹ Wilcoxon signed rank test for significant differences in growth before (1980–1985) and after (2010–2015) the outbreak: canopy ($V = 1892$, $p < 0.001$); subcanopy ($V = 8433$, $p < 0.001$); saplings ($V = 2350$, $p < 0.001$).

² Kruskal-Wallis tests for significant differences in growth among tree size classes: $X^2 = 19.2$, $df = 2$, $p < 0.0001$. Kolmogorov-Smirnov pairwise comparisons - canopy: subcanopy ($D = 0.19$, $p = 0.004$); canopy: saplings ($D = 0.15$, $p = 0.080$); subcanopy: saplings ($D = 0.17$, $p = 0.004$).

positive relationship between per cent reductions in canopy tree basal area and pre-outbreak basal area indicating that, on average, stands with the highest pre-outbreak tree basal area also had the highest per cent reductions in tree basal area after the beetle outbreak. Similarly, stands with the highest values for the climatic moisture deficit (i.e., stands with a drier climate) also had higher reductions in tree basal area following the outbreak.

None of the variables we tested could explain a significant amount

Table 4

Generalized linear models describing variability in (a) basal area reductions attributed to beetle outbreaks, (b) advance regeneration density, and (c) the percentage of trees with at least a 50% increase in growth, in relation to stand characters and climate. Statistically significant relationships are designated by *p*-values in bold text.

Dependant variable	Explanatory variable ¹	Parameters		<i>t</i> -value	<i>p</i>	Model		
		Estimate (β)	SE			<i>F</i>	<i>p</i>	Pseudo <i>R</i> ²
% basal area reduction by 2010	Intercept	−105.6	37.1	2.8	0.0112	11.9	0.0002	0.68
	Pre-outbreak tree basal area (m ² /ha)	1.1	0.3	3.6	0.0021			
	Climatic moisture deficit ²	0.5	0.2	2.7	0.0015			
	Humus depth (cm)	2.6	2.1	1.3	0.2243			
Advance regeneration density ³	All size classes	Intercept	19511.6	7344.5	2.7	0.0156	3.7	0.0697
		Climatic moisture deficit	−67.1	34.9	−1.9	0.0697		
	Saplings (0.3–1.3 m tall)	Intercept	444.9	1242.2	0.4	0.7242	4.6	0.0455
		Humus depth	544.4	254.3	2.1	0.0186		
	Small subcanopy trees (1.3–5 cm dbh)	Intercept	7449.0	2848.8	2.6	0.0170	4.1	0.0571
		Climatic moisture deficit	−27.4	13.5	−2.0	0.0571		
	Large subcanopy trees (5–10 cm dbh)	Intercept	115.3	4.2	4.2	< 0.0001	15.9	0.0001
		Pre-outbreak tree basal area (m ² /ha)	−0.4	0.1	−3.6	0.0018		
		Climatic moisture deficit	−0.3	0.1	−4.4	0.0004		
		Shrub cover (%)	−0.3	0.05	−4.5	0.0003		
	Proportion subcanopy trees With > 50% increased growth	Intercept	3.2	1.5	2.1	0.0328	15.1	< 0.0001
		Tree basal area, 2010	−0.1	0.01	−5.7	< 0.0001		
		Climate moisture deficit	−0.01	0.01	−2.3	0.0196		
		Humus depth	0.21	0.06	3.2	0.0010		

¹ All variables with a greater than zero effect, as determined by the LASSO procedure are included in the model.

² Climate moisture deficit (CMD) is a measure of the moisture needed for vegetation growth that must be met from sources other than rain (e.g., soil moisture) to avoid the impact of drought. It is the sum of the monthly difference between the Hargreaves reference evapotranspiration demand (Eref) and precipitation (mm) (see Wang et al., 2016). If precipitation is greater than Eref, CMD is zero and there is no moisture deficit. Increasing values of CMD indicate increasingly severe drought.

³ Trees establishing before the outbreak in 1994: saplings = 0.3–1.3 m tall; small subcanopy trees = 0–5 cm d.b.h.; large subcanopy trees = > 5–10 cm d.b.h.

of the variability in total advance regeneration (i.e., regeneration in all size classes combined) across stands; however, there were stronger relationships when analyses were conducted separately for each size class (Table 4). Large subcanopy trees were most sensitive to the variables we tested; inclusion of pre-outbreak tree basal area, climate moisture deficit, and shrub cover in the model reduced residual deviance by 63%. There was a significant negative relationship between all variables in the model and variability in large subcanopy tree density. Regression models for saplings and small subcanopy trees reduced residual deviance by only 19% and 18%, respectively. There was a positive relationship between humus depth and sapling density. There was a negative relationship, albeit of borderline statistical significance, between small subcanopy tree density and climate moisture deficit.

Three variables were included in the best logistic regression model describing post-outbreak stand variation in the proportion of surviving subcanopy trees with at least a 50% increase in radial growth: tree canopy basal area in 2010, climate moisture deficit, and humus depth (Table 4). These variables reduced residual deviance, compared to the null model, by 46%. There was a strong negative relationship between the proportion of subcanopy trees with at least a 50% increase in growth and tree basal area in 2010; growth releases were greatest for stands with the lowest canopy tree basal area. There was also a significant negative relationship between climate moisture deficit and the proportion of trees with increased growth. Humus depth had a positive relationship with variation in the proportion of trees with increased growth among stands.

4. Discussion

The recent outbreak of the spruce beetle caused a major reduction in the basal area and density of canopy trees in the spruce forests of southwestern Yukon. However, even in the stands most heavily affected by the spruce beetle outbreak, spruce remained a dominant component of the forest canopy (> 50%) in every stand at the end of the study. If depletion of hosts (canopy spruce) caused the beetle outbreak to end, we would have expected almost all canopy spruce to have died, as

occurred in spruce forests in both the southern parts of the beetle's geographic range and in Alaska (Berg et al., 2006; DeRose and Long, 2007; O'Connor et al., 2015; Hansen et al., 2016). Given that spruce beetles can, and often do, kill most host trees in the canopy, the moderately high survival of canopy spruce that we observed indicates that factors other than host depletion were involved in termination of the beetle outbreak. This is consistent with previous work showing bark beetle outbreaks, including spruce beetle outbreaks, are controlled by a variety of factors, climate and annual weather patterns being especially important (Raffa et al., 2008; Bentz et al., 2010; Weed et al., 2013; Jenkins et al., 2014; Temperli et al., 2015). In regions where bark beetle outbreaks do not deplete hosts, post-disturbance forest canopy recovery is greatly facilitated by the surviving trees.

We found high variability in outbreak severity among stands. While some boreal forests of Alaska exhibit similar patterns of spruce beetle outbreak severity (Doak, 2004; Werner et al., 2006), there was little variability in the severity of beetle outbreaks in Engelmann spruce forests of the western US as the beetle killed > 90% of trees in almost all stands (DeRose and Long, 2007; Bakaj et al., 2016). We found that pre-outbreak canopy tree basal area and the climatic moisture deficit explained a large percentage of among-stand variation in beetle-caused reductions in canopy tree basal area in southwestern Yukon, which is consistent with long-held views on stand-level effects (Werner et al., 2006). Productive stands with high tree basal area and many large spruce trees provide good habitat for spruce beetle population growth; the thick bark of large trees protects beetles from extreme winter temperatures and ample phloem provides habitat (food and brood space) for many beetles (Miller and Werner, 1987). Spruce trees weakened by drought are more susceptible to bark beetle attack (Hard and Holsten, 1985; Berg et al., 2006; Kolb et al., 2016) and this may explain why we found greater impacts of spruce beetle outbreaks among stands with the largest CMD values. In contrast to our findings, studies in western US indicate stand characteristics are not good predictors of the incidence of mortality from spruce beetle in subalpine forests (Hart et al., 2014; Bakaj et al., 2016). However, when beetle populations are high, stand-level constraints on damage are lessened, or removed, and

“the rules of the ecological game” are altered, with small trees, young trees, and even alternate hosts being attacked and killed by beetles.

Forests in many parts of western North America have recovered from past bark beetle outbreaks as advance regeneration rapidly grew to fill canopy gaps created by beetle-killed trees (Amoroso et al., 2013; Berg et al., 2006; Campbell et al., 2007; DeRose and Long, 2010; Hawkins et al., 2012; Pelz and Smith, 2012). We found very high densities of advance regeneration in the stands we studied, which is consistent with another recent study in Yukon (Paudel et al., 2015) and one in Colorado (Collins et al., 2011). Abundant regeneration, in addition to surviving canopy trees, as well as continued post-outbreak spruce recruitment (see supplement Fig. S2), indicate high potential for canopy recovery, even if some of the advance regeneration dies in the coming years. Greene et al. (2002) suggest ~38% of advance regeneration that establishes in boreal forests of Alberta will not survive, presumably due to high competition for resources. Despite such losses due to internal stand dynamics (Oliver and Larson, 1996), advance regeneration still plays a major role in forest recovery from bark beetle outbreaks and other major insect disturbances, such as eastern spruce budworm (Kneeshaw and Bergeron, 1998; MacLean, 2016), which is in sharp contrast to the new tree establishment required for recovery from severe forest fire disturbances (e.g., Paudel et al., 2015).

Spatial variability in the abundance of advance regeneration in boreal forests is related to many factors such as seed availability, forest canopy openness and competition intensity, humus depth, seedbed abundance, and herbivore abundance (Green et al., 1999; Solarik et al., 2010). Although substantial variability in advance regeneration density occurred among stands, we could only build a statistical model to adequately describe this variation for large subcanopy trees (> 5–10 cm dbh). Density of these large subcanopy trees increased with decreasing pre-canopy tree basal area and shrub cover, which is consistent with other studies of advance regeneration abundance in boreal forests (Kneeshaw and Bergeron, 1996; Green et al., 1999; Paudel et al., 2015); and suggests intense competition for above- or below-ground resources controls regeneration abundance in white spruce forests of Yukon. Stands with a higher climate moisture deficit index (i.e., more prone to drought) tended to have lower densities of advance regeneration but the strength of this relationship was weak for the smallest regeneration size classes. Others (Hogg and Wein, 2005; Paudel et al., 2015) found stronger relationships between regeneration abundance and drought in this region. The lack of, or weak relationship between small size classes of advance regeneration and the variables we tested – which were intended to capture climate effects, seedbed abundance, seed availability, and competition for above-ground and below-ground resources – suggests other factors are driving spatial variation in sapling and small canopy tree abundance among stands (e.g., hare browsing).

Even if advance regeneration density is adequate to form a new canopy, the species composition of this regeneration could lead to a very different future forest. In the forests we studied, white spruce dominated both the forest canopy and the advance regeneration providing no evidence of an imminent compositional shift. Paudel et al. (2015) studied the regeneration dynamics in our study area and report similar findings. However, several other studies report major shifts in forest composition following spruce beetle outbreaks. In the mountains of Alaska, outbreaks shifted forest composition to later successional species more tolerant of shade than spruce (i.e., *Tsuga mertensiana* (Bong.) Carrière, mountain hemlock) but in lowland regions, outbreaks generated dramatic shifts towards open forest communities with abundant early succession grasses and herbs (Boucher and Mead, 2006). In subalpine forests of southwestern US, DeRose and Long (2007) also found profound composition changes following spruce beetle outbreaks, with forests shifting from spruce to subalpine fir, aspen, or limber pine (*Pinus flexilis* James) dominance. While DeRose and Long (2007) report a major increase in aspen, we did not observe this in our study area even though aspen is an important species in these

boreal landscapes. Aspen basal area increased in our stands following the beetle outbreak but further increases are unlikely given that this shade-intolerant species is uncommon in the advance regeneration. Overall, our study indicated a high degree of compositional resilience to spruce beetle outbreaks because spruce remains the dominant species in both the advance regeneration and in the forest canopy, and because new establishment of shade-intolerant species is limited, again representing a sharp contrast to the effects of forest fires on forest composition (cf. Paudel et al., 2015).

The capacity of advance regeneration to grow faster in response to canopy thinning by beetle outbreaks is important for assessing the potential contribution of advance regeneration to canopy redevelopment. We found substantial growth increases in advance regeneration, which is consistent with results of other tree ring studies (Berg et al., 2006; Amoroso et al., 2013; Hawkins et al., 2013) and bodes well for the future contribution of advance regeneration to canopy recovery. It also reinforces the view that advance regeneration is likely to be a major source of new canopy trees following bark beetle disturbance (Nigh et al., 2008; Vyse et al., 2009; DeRose and Long, 2010; Campbell and Antos, 2015). However, we found that saplings were growing much slower, on average, than subcanopy trees and will take a much longer time to contribute to canopy recovery. Intensive browsing by hares – a common but a cyclical occurrence throughout the region (Krebs et al., 2001) – could increase the time some saplings spend in the understory (Olson and Kielland, 2016). But, not all saplings in our study were heavily browsed and the continuous tree establishment dates we observed in this study (unpublished data), suggest some small trees survive despite periods of intense herbivory. Given this, and the abundance of saplings we observed in most stands, we anticipate saplings could make an important long-term contribution to the canopy.

In addition to faster growing advance regeneration, we also found substantial increases in canopy tree growth that are contributing to the recovery of canopy basal area, as was observed in other regions following non-stand replacing insect disturbances (Coates et al., 2004; Amoroso et al., 2013). Release from lateral shading and intense competition for belowground soil resources (Canham et al., 2004; Coates et al., 2004) likely explain the increases in canopy tree growth we observed. We expected a changing climate, especially longer growing seasons, could also be contributing to the increases in the annual growth increment of canopy trees but our linear regression analyses indicated no significant relationship between the increasing tree growth trends and climate between 1998 and 2015, for any size class. Moreover, the temporal pattern of growth change we observed, particularly the stabilization of growth increment among canopy trees over the last decade, is consistent with growth releases from competition following canopy thinning during past beetle outbreaks (Berg et al., 2006; Campbell et al., 2007; Amoroso et al., 2013). While we anticipate canopy thinning due to bark beetle is the major cause of the growth increases we observed, more in-depth analyses are needed to tease apart the relative contributions of bark beetle outbreaks and climate change trends.

Although tree growth increased on average, we found substantial variability in post-beetle outbreak growth among trees within each size class and among stands. The spatial patterns of canopy tree mortality likely account for some of the observed variation in growth among trees of the same size class; some individuals may have lost all neighbours while others were surrounded by surviving trees. Many studies show neighbourhood competition to be a strong predictor of tree growth (Canham et al., 2004; Looney et al., 2016; Zhang et al., 2017; Fichtner et al., 2018). Among stands, variability in the incidence of subcanopy trees with at least a 50% increase in growth was related to tree canopy basal area, climatic moisture deficit, and humus depth underscoring the importance of resource availability to stand recovery potential. As reported in other post-disturbance studies (Donnegan and Rebertus, 1999; Brien et al., 2010; Campbell and Antos, 2015), we expect the potential and rates of stand recovery in southwest Yukon to vary along

gradients of resource availability. While we were unable to analyze variability in growth of canopy trees due to small sample sizes, we anticipate similar patterns among stands but possibly differences in response sensitivity (Coates et al., 2004).

Climate-induced changes in disturbance regimes, such as insect outbreaks, as well as changes in forest growth and regeneration, are expected to transform parts of the boreal forest into new states (Chapin et al., 2010; Johnstone et al., 2010; Boulanger et al., 2018). Several studies propose major state changes in western boreal forests, including a shift to hardwood-dominated landscapes (Wirth et al., 2008; Chapin et al., 2010; Johnstone et al., 2010; Paudel et al., 2015). Changes in stand development trajectories could trigger these transformations (Chapin et al., 2010). However, our results for southwestern Yukon suggest spruce forests are resilient to the last spruce beetle outbreak; bark beetles do not yet seem to be driving these systems towards an alternative state. The stands we sampled in southwestern Yukon currently possess a high capacity to develop a forest structure and composition that is similar to pre-outbreak forests. Many spruce survived the last outbreak, spruce dominates advance regeneration, spruce continues to establish, and we found no evidence of significant increases in hardwoods. Not only was white spruce advance regeneration common, its growth has increased substantially since the last outbreak providing additional evidence of resilience and suggesting a changed climate has not limited spruce growth in our study region. In contrast, studies in Alaska and other parts of western North America, indicate no change in growth (Trugman et al., 2018) or decreasing spruce growth due to recent climate change (Hogg et al., 2017).

Although the stands in our study are exhibiting high potential for recovery from the most recent bark beetle outbreak, the effects of future changes to disturbance regimes, compounded with the direct effects of a warming climate, may make these high-latitude ecosystems more vulnerable to changes in state (IPCC 2014; Reyher et al., 2015). We documented high inter-annual variability in the area infested by spruce beetle suggesting that beetles in this region were still near their cold tolerance thresholds during the last outbreak. If spruce forests maintain their dominance on the landscape, a subsequent outbreak could be more severe as the climate warms substantially over the latter part of the century (supplement Fig. S1). It is possible that continued lengthening of the growing seasons over the next few decades (supplement Fig. S1) could further accelerate spruce growth and stand recovery following beetle outbreaks. However, niche model projections for this region indicate major reductions in climatically suitable habitat for white spruce – and aspen, the other dominant trees species in this region – by the end of the century (supplement Fig. S3). Whether such changes could be driven by altered disturbance regimes – such as more severe beetle outbreaks or fires – or be the more direct effects of climate change (e.g., drought effects tree survival) remains highly uncertain.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2018.10.037>.

References

- Amoroso, M.M., Coates, K.D., Astrup, R., 2013. Stand recovery and self-organization following large-scale mountain pine beetle induced canopy mortality in northern forests. *For. Ecol. Manage.* 310, 300–311.
- Antos, J.A., Parish, R., Conley, K., 2000. Age structure and growth of the tree-seedling bank in subalpine spruce-fir forests of south-central British Columbia. *Am. Midl. Nat.* 143, 342–354.
- Aoki, C.F., Cook, M., Dunn, J., Finley, D., Fleming, L., Yoo, R., Ayres, M.A., 2018. Old pests in new places: effects of stands structure and forest type on susceptibility to a bark beetle on the edge of its range. *For. Ecol. Manage.* 419–420, 206–219.
- Astrup, R., Coates, K.D., Hall, E., 2008. Recruitment limitation in forests: lessons from an unprecedented mountain pine beetle epidemic. *For. Ecol. Manage.* 256, 1743–1750.
- Axelsson, J.N., Alfaro, R.I., Hawkes, B.C., 2009. Influence of fire and mountain pine beetle on the dynamics of lodgepole pine stands in British Columbia, Canada. *For. Ecol. Manage.* 257, 1874–1882.
- Bakaj, F., Mielkiewicz, N., Veblen, T.T., Kulakowski, D., 2016. The relative importance of tree and stand properties in susceptibility to spruce beetle outbreak in the mid-20th century. *Ecosphere* 7, 1–17. <https://doi.org/10.1002/ecs2.1485>.
- Bentz, B.J., Régnière, J., Fetting, C.J., Hansen, E.M., Hayes, J.L., Hicke, J.A., Kelsey, R.G., Negrón, J.F., Seybold, S.J., 2010. Climate change and bark beetles of the western United States and Canada: direct and indirect effects. *Bioscience* 60, 602–613. <https://doi.org/10.1525/bio.2010.60.8.6>.
- Berg, E.E., Henry, J.D., Fastie, C.L., De Volder, A.D., Matsuoka, S.M., 2006. Spruce beetle outbreaks on the Kenai Peninsula, Alaska, and Klauene National Park and Reserve, Yukon Territory: relationship to summer temperatures and regional differences in disturbance regimes. *For. Ecol. Manage.* 227, 219–232.
- Bergeron, Y., Flannigan, M., Gauthier, S., Leduc, A., Lefort, P., 2004. Past, current and future fire frequency in the Canadian boreal forest: implications for sustainable forest management. *AMBIO: J. Hum. Environ.* 33, 356–360.
- Boucher, T.V., Mead, B.R., 2006. Vegetation change and forest regeneration on the Kenai Peninsula, Alaska following a spruce beetle outbreak, 1987–2000. *For. Ecol. Manage.* 227, 233–246.
- Boulanger, Y., Taylor, A.R., Price, D.T., Cyr, D., Sainte-Marie, G., 2018. Stand-level drivers most important in determining boreal forest response to climate change. *J. Ecol.* 106, 977–990. <https://doi.org/10.1111/1365-2745.12892>.
- Brienen, R.J.W., Zuidema, P.A., Martínez-Ramos, M., 2010. Attaining the canopy in dry and moist tropical forests: strong differences in tree growth trajectories reflect variation in growing conditions. *Oecologia* 163, 485–496. <https://doi.org/10.1007/s00442-009-1540-5>.
- Campbell, E.M., Antos, J.A., 2015. Advance regeneration and trajectories of stand development following the mountain pine beetle outbreak in boreal forests of British Columbia. *Can. J. For. Res.* 45, 1327–1337.
- Campbell, E.M., Alfaro, R.I., Hawkes, B., 2007. Spatial distribution of mountain pine beetle outbreaks in relation to climate and stand characteristics: a dendroecological analysis. *J. Integr. Plant Biol.* 49, 168–178.
- Campbell, E.M., MacLean, D.A., Bergeron, Y., 2008. The severity of budworm-caused growth reductions in balsam fir/spruce stands varies with the hardwood content of surrounding forest landscapes. *For. Sci.* 54, 195–205.
- Candau, J.N., Fleming, R.A., Wang, X., 2018. Ecoregional patterns of spruce budworm-wildfire interactions in central Canada's forests. *Forests* 9, 137.
- Canham, C.D., LePage, P.T., Coates, K.D., 2004. A neighborhood analysis of canopy tree competition: effects of shading versus crowding. *Can. J. For. Res.* 34, 778–787.
- Chapin, F.S., Callaghan, T.V., Bergeron, Y., Fukuda, M., Johnstone, J.F., Juday, G., Zimov, S.A., 2003. Global change and the boreal forest: thresholds, shifting states or gradual change? *Ambio* 33 (6), 361–365.
- Chapin, F.S., McGuire, A.D., Ruess, R.W., Hollingsworth, T.N., Mack, M.C., Johnstone, J.F., Kasichke, E.S., Euskirchen, E.S., Jones, J.B., Jorgenson, M.T., Kielland, K., Kofinas, G.P., Turetsky, M.R., Yarie, J., Lloyd, A.H., Taylor, D.L., 2010. Resilience of Alaska's boreal forest to climatic change. *Can. J. For. Res.* 40, 1360–1370. <https://doi.org/10.1139/X10-074>.
- Chavardes, R.D., Daniels, L.D., Waeber, P.O., Innes, J.L., Nischke, C.R., 2012. Did the 1976–77 switch in the Pacific Decadal Oscillation make white spruce in the south-west Yukon more susceptible to spruce bark beetle? *For. Chron.* 88, 513–518.
- Coates, K.D., Canham, C.D., Beaudet, M., Sachs, D.L., Messier, C., 2004. Use of a spatially explicit individual-tree model (SORTIE/BC) to explore the implications of patchiness in structurally complex forests. *For. Ecol. Manage.* 186, 297–310.
- Collins, B.J., Rhoades, C.C., Hubbard, R.M., Battaglia, M.A., 2011. Tree regeneration and future stand development after bark beetle infestation and harvesting in Colorado lodgepole pine stands. *For. Ecol. Manage.* 261, 2186–2186-2175.
- Csank, A.Z., Miller, A.E., Sherriff, R.L., Berg, E.E., Welker, J.M., 2016. Tree-ring isotopes reveal drought sensitivity in trees killed by spruce beetle outbreaks in south-central Alaska. *Ecol. Appl.* 26, 2001–2020.
- Cullingham, C.I., Cooke, J.E.K., Dang, S., Davis, C.S., Cooke, B.J., Coltman, D.W., 2011. Mountain pine beetle host-range expansion threatens the boreal forest. *Mol. Ecol.* 20, 2157–2171.
- Dalhgren, J.P., 2010. Alternative regression methods are not considered in Murtaugh (2009) or by ecologists in general. *Ecol. Lett.* 13, E7–E9.
- de Groot, W.J., Flannigan, M.D., Cantin, A.S., 2013. Climate change impacts on future

- boreal fire regimes. *For. Ecol. Manage.* 294, 35–44. <https://doi.org/10.1016/j.foreco.2012.09.027>.
- DeRose, R.J., Long, J.N., 2007. Disturbance, structure, and composition: spruce beetle and Engelmann spruce forest on the Markagunt Plateau, Utah. *For. Ecol. Manage.* 244, 16–23.
- DeRose, R.J., Long, J.N., 2010. Regeneration response and seedling bank dynamics on a *Dendroctonus rufipennis*-killed *Picea engelmannii* landscape. *J. Veg. Sci.* 21, 377–387.
- DeRose, R.J., Long, J.N., 2012. Factors influencing the spatial and temporal dynamics of Engelmann spruce mortality during a spruce beetle outbreak on the Markagunt Plateau, Utah. *For. Sci.* 58, 1–14.
- Doak, P., 2004. The impact of tree and stand characteristics on spruce beetle (Coleoptera: Scolytidae) induced mortality of white spruce in the Copper River Basin, Alaska. *Can. J. For. Res.* 34, 810–816.
- Donnegan, J.A., Rebertus, A.J., 1999. Rates and mechanisms of subalpine forest succession along an environmental gradient. *Ecology* 80, 1370–1384.
- Eisenhart, K.S., Veblen, T.T., 2000. Dendroecological detection of spruce bark beetle outbreaks in northwestern Colorado. *Can. J. For. Res.* 30, 1788–1798.
- Fichtner, A., Härdtle, W., Bruehlheide, H., Kunz, M., Li, Y., von Oheimb, G., 2018. Neighbourhood interactions drive overyielding in mixed-species tree communities. *Nat. Commun.* 9, 1144. <https://doi.org/10.1038/s41467-018-03529-w>.
- Francis, R.R., 1996. Linking Landscape Pattern and Forest Disturbance: Fire History of the Shikwak Trench, Southwest Yukon Territory (M.Sc. Thesis). University of Alberta, Edmonton. <https://doi.org/10.7939/R37940Z9J>.
- Franklin, J.F., Mitchell, R.J., Palik, B., 2007. Natural disturbance and stand development principles for ecological forestry. USDA For. Serv. Gen. Tech. Rep. NRS-19, 48p.
- Friedman, J., Hastie, T., Tibshirani, R., Simon, Noah, Narasimhan, B., Qian, J., 2018. R Package 'glmnet'. Available: < <http://cran.r-project.org/web/packages/glmnet/index.html> > (accessed 2018 August 25).
- Fuentealba, A., Alfaro, R., Bauce, E., 2013. Theoretical framework for assessment of risks posed to Canadian forests by invasive insect species. *For. Ecol. Manage.* 302, 97–106.
- Garbutt, R., Hawkes, B., Allen, E., 2006. Spruce beetle and the forests of the southwest Yukon. Information Report BC-X-406, Canadian Forest Service, Pacific Forestry Centre.
- Gauthier, S., Bernier, P., Kuuluvainen, T., Shvidenko, A.Z., Schepaschenko, D.G., 2015. Boreal forest health and global change. *Science* 349, 819–822. <https://doi.org/10.1126/science.aaa9092>.
- Greene, D.F., Kneeshaw, D.D., Messier, C., Lieffers, V., Cormier, D., Doucet, R., Coates, K.D., Groot, A., Grover, G., Calogeropoulos, C., 2002. Modelling silviculture alternatives for conifer regeneration in boreal mixedwoods (aspen/white spruce/balsam fir). *For. Chron.* 78, 281–295.
- Green, D., Zasda, J., Sirois, L., Kneeshaw, D., Morin, H., Charron, I., Simard, M.-J., 1999. A review of the regeneration dynamics of North American boreal forest tree species. *Can. J. For. Res.* 29, 824–839.
- Hansen, W.D., Chapin III, F.S., Naughton, H.T., Rupp, T.S., Verbyla, D., 2016. Forest-landscape structure mediates effects of a spruce bark beetle (*Dendroctonus rufipennis*) outbreak on subsequent likelihood of burning in an Alaskan boreal forest. *For. Ecol. Manage.* 369, 38–46.
- Hard, J.S., Holsten, E.H., 1985. Managing white and Lutz spruce stands in south-central Alaska for increased resistance to spruce beetle. USDA For. Serv. Gen. Tech. Rep. PNW-GTR-188 21p. <https://doi.org/10.2737/PNW-GTR-188>.
- Hart, S.J., Veblen, T.T., Eisenhart, K.S., Jarvis, D., Kulakowski, D., 2014. Drought induces spruce beetle (*Dendroctonus rufipennis*) outbreaks across northwestern Colorado. *Ecology* 95, 930–939.
- Hawkes, B., 1983. Fire history and management study of Kluane National Park. In: Methven, I.R., Riewe, R.R., Wein, R.W. (Eds.), *Resources and Dynamics of the Boreal Zone: Proceedings of a Conference Held at Thunder Bay, Ontario, August 1982*. Association of Canadian Universities for Northern Studies, Ottawa.
- Hawkins, C.D.B., Dhar, A., Balliet, N.A., Runzer, K.D., 2012. Residual mature trees and secondary stand structure after mountain pine beetle attack in central British Columbia. *For. Ecol. Manage.* 277, 107–115.
- Hawkins, C.D.B., Dhar, A., Balliet, N.A., 2013. Radial growth of residual canopy trees and understory saplings after mountain pine beetle attack in central British Columbia. *For. Ecol. Manage.* 310, 348–356.
- Hogg, E.H., Wein, R.W., 2005. Impacts of drought on forest growth and regeneration following fire in southwestern Yukon, Canada. *Can. J. For. Res.* 35, 2141–2150.
- Hogg, E.H., Michaelian, M., Hook, T.I., Undersultz, M.E., 2017. Recent climatic drying leads to age-independent growth reductions of white spruce stands in western Canada. *Glob. Change Biol.* 23, 5297–5308. <https://doi.org/10.1111/gcb.13795>.
- Holmes, R.L., 1983. Computer-assisted quality control in tree-ring dating and measurement. *Tree-Ring Bull.* 43, 69–78.
- IPCC, 2014. Summary for policymakers. Climate change 2014. Impacts, adaptation, and vulnerability. Part A: global and sectoral aspects. In: Field, C.B., Barros, V.R., Kokken, D.J., Mach, K.J., Mastrandrea, M.D., Bilir, T.E., Chatterjee, M., Ebi, K.L., Estrada, Y.O., Genova, R.C., Girma, B., Kissel, E.S., Levy, A.N., MacGracken, D.J., Mastrandrea, P.R., White, L.L. (Eds.), *Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press, Cambridge, UK and New York, NY, USA, pp. 32.
- Iverson, L.R., Perera, A.H., Pastur, G.M., Peterson, U., 2018. Ecosystem services from forest landscapes: where we are and where we go. In: Perera, A., Peterson, U., Pastur, G., Iverson, L. (Eds.), *Ecosystem Services from Forest Landscapes*. Springer, Cham, pp. 249–258. <http://doi.org/10.1007/978-3-319-74515-2>.
- Jenkins, M.J., Herberston, E.G., Munson, A.S., 2014. Spruce beetle biology, ecology, and management in the Rocky Mountains: an addendum to Spruce Beetle in the Rockies. *Forests* 5, 21–71.
- Johnson, E.A., 1992. Fire and Vegetation Dynamics: Studies from the North American Boreal Forest. Cambridge University Press, Cambridge, UK, pp. 129.
- Johnstone, J.F., Chapin, F.S., Hollingsworth, T.N., Mack, M.C., Romanovsky, V., Turetsky, M., 2010. Fire, climate change, and forest resilience in interior Alaska. *Can. J. For. Res.* 40, 1302–1312. <https://doi.org/10.1139/X10-061>.
- Kneeshaw, D.D., Bergeron, Y., 1998. Ecological factors affecting the abundance of advance regeneration in Quebec's southwestern boreal forest. *Can. J. For. Res.* 26, 888–898.
- Kneeshaw, D.D., Bergeron, Y., 1996. Canopy gap characteristics and tree replacement in the southeastern boreal forest. *Ecology* 79, 783–794.
- Kolb, T.E., Fettig, C.J., Ayres, M.P., Bentz, B.J., Hicke, J.A., Mathiasen, R., Stewart, J.E., Weed, A.S., 2016. Observed and anticipated impacts of drought on forests insects and diseases in the United States. *For. Ecol. Manage.* 380, 321–334.
- Krebs, C.J., Boutin, S., Boonstra, R., 2001. *Ecosystem Dynamics of the Boreal forest: Kluane Project*. Oxford University Press, pp. 544.
- Levin, S.A., 2005. Self-organization and the emergence of complexity in ecological systems. *Bioscience* 55, 1075–1079. [https://doi.org/10.1641/0006-3568\(2005\)055\[1075:SATEOC\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2005)055[1075:SATEOC]2.0.CO;2).
- Lewkowicz, A.G., Bonnaventure, P.P., Smith, S.L., Kuntz, Z., 2012. Spatial and thermal characteristics of mountain permafrost, northwest Canada. *Geogr. Ann.* 94, 195–213.
- Looney, C.E., D'Amato, A.W., Fraver, S., Palik, B.J., Reinikainen, M.R., 2016. Examining the influences of tree-to-tree competition and climate on size-growth relationships in hydric, multi-aged *Fraxinus nigra* stands. *For. Ecol. Manage.* 375, 238–248.
- MacLean, D.A., 2016. Impacts of insect outbreaks on tree mortality, productivity and stand development. *Can. Ent.* 148 (S1), S138–S159. <https://doi.org/10.4039/tce.2015.24>.
- Millar, C.I., Stephenson, N.L., 2015. Temperate forest health in an era of emerging megadisturbance. *Science* 349, 823–826.
- Miller, T.E., Werner, P.A., 1987. Cold-hardiness of adult and larval spruce beetle *Dendroctonus rufipennis* (Kirby) in interior Alaska. *Can. J. Zool.* 65, 2927–2930.
- Morris, J.L., Cottrell, S., Fettig, C.J., Hansen, W.D., Sherriff, R.L., Carter, V.A., Clear, J.L., Clement, J., DeRose, R.J., Hicke, J.A., Higuera, P.E., Mattor, K.M., Seddon, A.W.R., Seppä, H.T., Stednick, J.D., Seybold, S.J., 2017. Managing bark beetle impacts on ecosystems and society: priority questions to motivate future research. *J. Appl. Ecol.* 54, 750–760.
- Muggeo, V., 2017. R Package 'segmented'. Available: < <http://cran.r-project.org/web/packages/segmented/index.html> > (accessed 2018 August 25).
- Nigh, G.D., Antos, J.A., Parish, R., 2008. Density and distribution of advance regeneration in mountain pine beetle killed lodgepole pine stands of the Montane Spruce zone of southern British Columbia. *Can. J. For. Res.* 38, 2826–2836.
- O'Connor, C.D., Lynch, A.M., Falk, D.A., Swetnam, T.W., 2015. Post-fire forest dynamics and climate variability affect spatial and temporal properties of spruce beetle outbreaks on a Sky Island mountain range. *For. Ecol. Manage.* 336, 148–162.
- Oliver, C.D., Larson, B., 1996. *Forest Stand Dynamics*. John Wiley & Sons Inc, USA, pp. 520.
- Olson, J., Kielland, K., 2016. Stage-dependent effects of browsing by snowshoe hares on successional dynamics in a boreal forest ecosystem. *Ecosphere* 7 (10), e01475. <https://doi.org/10.1002/ecs2.1475>.
- Paudel, S.K., Simard, S.W., Nitschke, C.R., Innes, J.L., 2015. Climate variation and disturbance regime affect stand composition and structure of the boreal forests in southwest Yukon of Canada. *Open J. For. S.* 5, 337–352.
- Payette, S., 1992. Fire as a controlling process in the North American boreal forest. In: Shugart, H.H., Leemans, R., Bonan, G.B. (Eds.), *A Systems Analysis of the Global Boreal Forest*. Cambridge University Press, Cambridge, UK.
- Pelz, K.A., Smith, F.W., 2013. How will aspen respond to mountain pine beetle? A review of literature and discussion of knowledge gaps. *For. Ecol. Manage.* 299, 60–69.
- Pelz, K.A., Smith, F.W., 2012. Thirty year change in lodgepole and lodgepole/mixed conifer forest structure following 1980s mountain pine beetle outbreak in western Colorado, USA. *For. Ecol. Manage.* 280, 93–102.
- Peutmann, K., Coates, K.D., Messier, C., 2008. *A Critique of Silviculture; Managing for Complexity*. Island Press, Washington, D.C.
- Price, D.T., Alfaro, R.I., Brown, K.J., Flannigan, M.D., Fleming, R.A., Hogg, E.H., Girardin, M.P., Lakusta, T., Johnston, M., McKenney, D.W., Padlar, J.H., Stratton, T., Surrock, R.N., Thompson, I.D., Trofymow, J.A., Venier, L.A., 2013. Anticipating the consequences of climate change for Canada's boreal forest ecosystems. *Environ. Rev.* 21, 322–365.
- R Development Core Team, 2016. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. Available: < <http://www.R-project.org> > (accessed May 2018).
- Raffa, K.F., Aukema, B.H., Bentz, B.J., Carroll, A.L., Hicke, J.A., Turner, M.G., Romme, W.H., 2008. Cross-scale drivers of natural disturbances prone to anthropogenic amplification: the dynamics of bark beetle eruptions. *Bioscience* 58, 501–517.
- Reyer, C.S., Rammig, A., Brouwers, N., Langerwisch, F., 2015. Forest resilience, tipping points and global change processes. *J. Ecol.* 103, 1–4.
- Scheffer, M., Hirota, M., Holmgren, M., Van Nes, E.H., Chapin, F.S., 2012. Thresholds for boreal biome transitions. *Proc. Natl. Acad. Sci. USA* 109, 21384–21389.
- Seidl, R.S., Donato, D.C., Raffa, K.F., Turner, M.G., 2016. Spatial variability in tree regeneration after wildfire delays and dampens future bark beetle outbreaks. *Proc. Natl. Acad. Sci. USA* 113, 13075–13080.
- Seidl, R.S., Müller, J., Hothorn, T., Bässler, C., Heurich, M., Kautz, M., 2015. Small beetle, large-scale drivers: how regional and landscape factors affect outbreaks of the European spruce bark beetle. *J. Appl. Ecol.* 53, 530–540.
- Shorohova, E., Kneeshaw, D., Kuuluvainen, T., Gauthier, S., 2011. Variability and dynamics of old-growth forests in the circumboreal zone: implications for conservation, restoration and management. *Silva Fennica* 45, 72. <https://doi.org/10.14214/sf.72>.
- Solarik, K., Lieffers, V., Volney, W., Pelletier, R., Spence, J., 2010. Seed tree density, variable retention, and stand composition influence the recruitment of white spruce

- in boreal mixedwood forests. *Can. J. For. Res.* 40, 1821–1832.
- Swetnam, T.W., Thompson, M.A., Sutherland, E.K., 1985. Using dendrochronology to measure radial growth of defoliated trees. *USDA For. Serv. Agriculture Handbook* 639. 39pp.
- Temperli, C., Veblen, T.T., Hart, S.J., Kulakowski, D., Tepley, A., 2015. Interactions among spruce beetle disturbance, climate change and forest dynamics captured by a forest landscape model. *Ecosphere* 6, 231. <https://doi.org/10.1890/ES15-00394.1>.
- Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. *J. Roy. Stat. Soc. Ser. B* 58, 267–288.
- Trugman, A.T., Medvigy, D., Anderegg, W.R.L., Pacala, S.W., 2018. Differential declines in Alaskan boreal forest vitality related to climate and competition. *Glob. Change Biol.* 24, 1097–1107. <https://doi.org/10.1111/gcb.13952>.
- Veblen, T.T., Hadley, K.S., Reid, M.S., Rebertus, A.J., 1991. The response of subalpine forests to spruce beetle outbreak in Colorado. *Ecology* 72, 213–231.
- Vyse, A., Ferguson, C., Huggard, D.J., Roach, J., Zimonick, B., 2009. Regeneration beneath lodgepole pine dominated stands attacked or threatened by the mountain pine beetle in the south central interior, British Columbia. *For. Ecol. Manage.* 258S, S36–S43.
- Wang, T., Hamann, A., Spittlehouse, D.L., Carroll, C., 2016. Locally downscaled and spatially customizable climate data for historical and future periods for North America. *PLOS One* 11 (6), e0156720. <https://doi.org/10.1371/journal.pone.0156720>.
- Weed, A.S., Ayres, M.P., Hicke, J.A., 2013. Consequences of climate change for biotic disturbances in North American forests. *Ecol. Monogr.* 83, 441–470.
- Werner, R.A., Holsten, E.H., Matsuoka, S.M., Burnside, R.E., 2006. Spruce beetles and forest ecosystems in south-central Alaska: a review of 30 years of research. *For. Ecol. Manage.* 227, 195–206.
- Wirth, R., Meyer, S.T., Leal, I.R., Tabarelli, M., 2008. Plant–herbivore interactions at the forest edge. *Prog. Bot.* 69, 423–448.
- Yamaguchi, D.K., 1991. A simple method of cross-dating increment cores from living trees. *Can. J. For. Res.* 21, 414–416.
- Yukon Government, 2002–2017. Yukon Forest Health Reports. Government of Yukon, Energy, Mines and Resources. Retrieved May, 2018. < http://www.emr.gov.yk.ca/forestry/forest_health_reports.html > .
- Yukon Government, 1994–2014. Forest Health Aerial Overview. Forest Management Branch, Department of Energy, Mines, and Resources. Retrieved March, 2017. < ftp://ftp.geomaticsyukon.ca/GeoYukon/Forestry/Forest_Health_Aerial_Overview_50k/ > .
- Zhang, Q., Alfaro, R.I., Hebda, R.J., 1999. Dendroecological studies of tree growth, climate and spruce beetle outbreaks in Central British Columbia, Canada. *For. Ecol. Manage.* 121, 215–225.
- Zhang, Z., Papaik, M.J., Wang, X., Hao, Z., Ye, J., Lin, F., Yuan, Z., 2017. The effect of tree size, neighborhood competition and environment on tree growth in an old-growth temperate forest. *J. Plant Ecol.* 10, 970–980. <https://doi.org/10.1093/jpe/rtw126>.